Contents

0.1 Probability Spaces

Definition 0.1.1 ▶ Probability Spaces

A ***probability space*** is a measure space $(\Omega, \mathcal{F}, P)$ with

$$P : \mathcal{F} \to [0, 1]$$

Example 0.1.2 ▶ Discrete probability spaces

Let Ω be a countable set . Let $\mathcal{F}$ be the collection of all subsets of Ω . Let

$$P(A) = \sum_{\omega \in A} p(\omega), \quad \text{where } p(\omega) \geq 0 \text{ and } \sum_{\omega \in \Omega} p(\omega) = 1$$